# SOLUÇÕES FICHA 3

## B. INTEGRAL DE RIEMANN EM $\mathbb{R}$

### B.1. Conceito de integral definido

**3.1.**

**a)** $30$ **b)** $26$

**c)** $1/3$ **d)** $-55/6$

**e)** $21$ **f)** $28/3$

**g)** $17/3$ **h)** $2\sqrt{5}-2\sqrt{2}$

**i)** $\sqrt[5]{5}\left[\frac{5}{6}-\frac{10}{3}\sqrt[5]{4}\right]$ **j)** $21\sqrt[3]{7}$

**k)** $-4/3$ **l)** $28/3$

**3.2.**

**a)** $5/6$ **b)** $\dfrac{c^2+c-1}{2}+c-c^2$

**3.3.**

$P(x)=6x(1-x)$

**3.4.**

**a)** $\int_{-b}^{b} f(x)\,dx=0$ **b)** $\int_{-b}^{b} f(x)\,dx=2\int_{0}^{b} f(x)\,dx$

**3.5.**

**a)** $1-\cos(p/2)\geq 0$ **b)** $\cos(p/2)-\cos(p/2)=0$

## B.2. Cálculo de áreas e teoremas do valor médio para integrais

**3.6.**

$28/3$

**3.7.**

**a)** $4/3$ **b)** $2\frac{\sqrt{2}}{3}$

**c)** $\frac{14-4\sqrt{8}}{6}$ **d)** $7/3$

**e)** $5$ **f)** $\frac{5\sqrt{5}-7}{6}$

**g)** $32/3$

**3.8.**

**a)** $x_1 = \frac{10}{6}, x_2 = -1$ **b)** $x = \left(\frac{45}{28}\right)^3$

**3.9.**

**a)** $x = \arccos\left(\frac{2}{\pi}\right)$ **b)** $x = \frac{\pi}{4}$

## B.3. Teoremas fundamentais do cálculo

**3.10.**

**a)** $F(x) = \frac{4x^3}{3} + \frac{5x^2}{2} + 6x$ **b)** $F(x) = \frac{1}{2}(\sin^2 x - 1)$

**c)** $F(x) = \frac{x^2}{2} + x$ **d)** $F(x) = \frac{1}{2}(1 + x^2)$

**3.11.**

$f(x) = 2x + 2\cos(x) - \sin(2x)$ $f(\pi/2) = (\pi - 2)$ $f'(\pi/4) = -2$

**3.12.**

**a)** $F'(x) = f(x) = (-2) \cdot \sin(2x), c = \frac{\pi}{6}$ **b)** $F'(x) = f(x) = x \cdot \sin(x) - x, c = 0$

**c)** $F'(x) = f(x) = \frac{2x^{15} + 2x^{17}}{(1+x^2)}, c = -\frac{1}{9}$

**3.13.**

**a)** $F'(x) = \left(1+x^2\right)^{-3}$ **b)** $F'(x) = 2x\left(1+x^4\right)^{-3}$

**c)** $F'(x) = 2x\left(1+x^4\right)^{-3} - 3x^2\left(1+x^6\right)^{-3}$

**3.14.**

**a)** $f\,(2) = 16$ **b)** $f\,(2) = 1 + \frac{3\sqrt{2}}{2}$

**c)** $f\,(2) = \sqrt[3]{36}$ **d)** $f\,(2) = \frac{1}{5}$

**3.15.**

$h(x) = -\frac{1}{2}\ln|1 + \cos(x)| + C$

## B.4. Primitivação por substituição e por partes

**3.16.**

$\int f(x)\,dx = \frac{1}{2}|x|x + C$

**3.17.**

$f(x) = \begin{cases} 2 - e^{-x}, & x < 0 \\ e^{x}, & x \geq 0 \end{cases}$

**3.18.**

**a)** $P(x) = \frac{2}{3}(x+1)^{3/2} + C$

**b)** $P(x) = \frac{1}{2}\arctan\left(\frac{x}{2}\right) + C$

**c)** $P(x) = \frac{3}{4}\sqrt[3]{(1+2e^x)^2} + C$

**d)** $P(x) = x + \frac{5}{2}\ln|x^2+4| + \arctan\left(\frac{x}{2}\right) + C$

**e)** $P(x) = \frac{a^x}{\ln a} + C$

**f)** $P(x) = \arcsin\left(\frac{x}{\sqrt{2}}\right) + C$

**g)** $P(x) = \frac{2^{3x}}{3\ln 2} + C$

**h)** $P(x) = \frac{1}{2}\tan(x^2) + C$

**i)** $P(x) = \frac{\ln 10}{2}[\log_{10}(x)]^2 + C$

**j)** $P(x) = \ln|\sin(x)| + C$

**k)** $P(x) = -\frac{1}{(1+2x)^2} + C$

**l)** $P(x) = \frac{\sin^3(x)}{3} + C$

**m)** $P(x) = \frac{2a}{(a-x)} + C$

**n)** $P(x) = \frac{1}{2}\ln\left|e^{x^2-1} - 1\right| + C$

**o)** $P(x) = \frac{1}{2a}(\ln|a+x| - \ln|a-x|) + C$

**p)** $P(x) = \frac{1}{3}\sin(x^3) + C$

**q)** $P(x) = \frac{1}{4}\ln|x^4 + a^4| + C$

**r)** $P(x) = \frac{1}{2}\sec(2x) + C$

**s)** $P(x) = \frac{a+bx}{b^2} - \frac{a}{b^2}\ln|a+bx| + C$

**t)** $P(x) = \dfrac{e^x - e^{-x}}{2} + C$

**u)** $P(x) = \dfrac{(x-1)^2}{2} + 2(x-1) + 2\ln|x-1| + C$

**v)** $P(x) = -\dfrac{e^{\cos^2(x)}}{2} + C$

**x)** $P(x) = \ln|x+1| + \dfrac{1}{x+1} + C$

**y)** $P(x) = \dfrac{1}{2}[\ln(x)]^2 + C$

**3.19.**

**a)** $P(x) = \dfrac{e^{x\cdot\ln(3)+x}}{\ln(3)+1} - \dfrac{1}{\ln(3)+1}$ **b)** $P(x) = \dfrac{1}{3}\left(e^{x^3} - 1\right)$

**c)** $P(x) = \arcsin(e^x) - \dfrac{\pi}{2}$

**3.20.**

**a)** $P(x) = -e^{1/x} + C$

**b)** $P(x) = \dfrac{16^{x^2-1}}{\ln(16)} + C$

**c)** $P(x) = \sqrt{x^2+1} + C$

**d)** $P(x) = \dfrac{2}{45}\sqrt{(1+3x)^5} - \dfrac{2}{27}\sqrt{(1+3x)^3} + C$

**e)** $P(x) = \dfrac{2}{7}\sqrt{(1+x)^7} - \dfrac{4}{5}\sqrt{(1+x)^5} + \dfrac{2}{3}\sqrt{(1+x)^3} + C$

**f)** $P(x) = -\dfrac{4}{9}\sqrt{2-3x} + \dfrac{2}{27}\sqrt{(2-3x)^3} + C$

**g)** $P(x) = \dfrac{4}{9}\sqrt[4]{(x+1)^9} - \dfrac{4}{5}\sqrt[4]{(x+1)^5} + C$

**h)** $P(x) = -\dfrac{1}{3}\sqrt{1-x^6} + C$

**i)** $P(x) = \sqrt{x^2+2x+3} + C$

**j)** $P(x) = \dfrac{1}{2}\ln|1+x^2| - \ln\left|1+\sqrt{1+x^2}\right| + C$

**k)** $P(x) = \dfrac{1}{3 + \cos(x)} + C$

**l)** $P(x) = -2 \cdot \cos\left(\sqrt{x+1}\right) + C$

**m)** $P(x) = \cos\left(\dfrac{1}{x}\right) + C$

**n)** $P(x) = -\dfrac{2}{9}\sqrt{[1 + 3\cos^2(x)]^3} + C$

**3.21.**

**a)** $P(x) = x \cdot \sin(x) + \cos(x) + C$

**b)** $P(x) = (-x^2 + 2) \cdot \cos(x) + 2x \cdot \sin(x) + C$

**c)** $P(x) = -\dfrac{1}{4} \cdot \sin(2x) + \dfrac{x}{2} + C$

**d)** $P(x) = \dfrac{\sin(x) \cdot \cos^4(x)}{5} + \dfrac{4}{5} \cdot \sin(x) - \dfrac{4}{15} \cdot \sin^3(x) + C$

**e)** $P(x) = \dfrac{\sin(x) \cdot \cos^3(x)}{4} + \dfrac{3}{8} \cdot [\sin(x) \cdot \cos(x) + x] + C$

**f)** $P(x) = \dfrac{e^x}{2} \cdot [\cos(x) + \sin(x)] + C$

**g)** $P(x) = e^x \cdot (x^2 - 2x + 2) + C$

**h)** $P(x) = x \cdot \log_{10}(1 + x^2) - \dfrac{2}{\ln 10} \cdot [x - \arctan(x)] + C$

**i)** $P(x) = \dfrac{x^2}{2} \cdot [\log_{10}(x)]^2 - \dfrac{x^2}{2} \cdot \dfrac{\log_{10}(x)}{\ln 10} + \dfrac{x^2}{4 \cdot (\ln 10)^2} + C$

**j)** $P(x) = \dfrac{x^3}{3} \cdot \log_{10}(1 + x) - \dfrac{x^3}{9 \cdot \ln 10} + \dfrac{x^2}{6 \cdot \ln 10} - \dfrac{x}{3 \cdot \ln 10} + \dfrac{1}{3 \cdot \ln 10} \cdot \log_{10}(1 + x) + C$

**k)** $P(x) = x \cdot \arcsin\left(\dfrac{x}{\sqrt{2}}\right) + \sqrt{2} \cdot \sqrt{1 - \dfrac{x^2}{2}} + C$

**l)** $P(x) = \dfrac{x^2}{2} \cdot \arctan(x) - \dfrac{x}{2} + \dfrac{1}{2} \cdot \arctan(x) + C$

**m)** $P(x) = \sqrt{1 + x^2} \cdot \arctan(x) - \ln\left|\sqrt{1 + x^2} + x\right| + C$

**n)** $P(x) = x^2 \cdot \sqrt{1 + x^2} - \dfrac{2}{3} \cdot (1 + x^2)^{\frac{3}{2}} + C$

**o)** $P(x) = \dfrac{-x \cdot \sqrt{1 - x^2} + \arcsin(x)}{2} + C$

**p)** $P(x) = \frac{x}{8} - \frac{\sin(4x)}{32} + C$

**q)** $P(x) = x \cdot \tan(x) + \ln|\cos(x)| + C$

**r)** $P(x) = \frac{1}{2} \cdot \sec(x) \cdot \tan(x) + \frac{1}{2} \cdot \ln|\sec(x) + \tan(x)| + C$

**s)** $P(x) = \frac{\tan^3(x)}{3} - \tan(x) + x + C$

**t)** $P(x) = -\frac{\cot^4(x)}{4} + \frac{\cot^2(x)}{2} + \ln|\sin(x)| + C$

**u)** $P(x) = x \cdot \ln(1 + x^2) - 2 \cdot [x - \arctan(x)] + C$

**v)** $P(x) = \frac{x^2}{2} \cdot [\ln(x)]^2 - \frac{x^2}{2} \cdot \ln(x) + \frac{x^2}{4} + C$

**w)** $P(x) = \frac{x^3}{3} \cdot \ln(1 + x) - \frac{x^3}{9} + \frac{x^2}{6} - \frac{x}{3} + \frac{1}{3} \cdot \ln(1 + x) + C$

## B.5. Cálculo de volumes usando integrais

**3.22.**

**a)** $\frac{2}{3}\pi\ (u.v.)$

**b)** $\frac{117}{5}\pi\ (u.v.)$

**c)** $\frac{2}{3}\pi\ (u.v.)$

**d)** $\frac{64}{15}\pi\ (u.v.)$

**e)** $---\ (u.v.)$

**3.23.**

**a)** $\frac{256}{5}\pi\ (u.v.)$

**b)** $\frac{512}{15}\pi\ (u.v.)$

**c)** $\frac{832}{15}\pi\ (u.v.)$

**d)** $\frac{128}{3}\pi\ (u.v.)$

**e)** $64\pi\ (u.v.)$

**3.24.**

**a)** $\frac{17}{2}\pi\ (u.v.)$

**b)** $---$

**c)** $90\pi\ (u.v.)$

**d)** $-2\sqrt{13}\pi + \frac{253\sqrt{13}\pi + 521\pi}{120} \approx 14.87\ (u.v.)$

**e)** $\frac{\pi^2}{16}\ (u.v.)$

## B.6. Definição de funções e cálculo de áreas usando coordenadas polares

**3.25.**

**a)** $r = \sqrt{\cos(2\theta)} + \cos(\theta)$

**b)** $r = \frac{6}{\sqrt{9 - 5 \cdot \sin^2(\theta)}}$

**c)** $r = 2 \cdot \cos(\theta)$

**d)** $r = \sqrt{|\cos(2\theta)|}$

**e)** $r = \sqrt{\cos(2\theta)}$

**Nota.** Soluções com $r$ considerado sempre positivo.

**3.26.**

**a)** $\frac{4}{3}\pi^3\ (u.a.)$

**b)** $\pi\ (u.a.)$

**c)** $2\pi\ (u.a.)$

**d)** $\frac{\pi}{8}\ (u.a.)$

**e)** $\frac{\pi}{2}\ (u.a.)$

**f)** $4\pi\ (u.a.)$

**g)** $8\pi\ (u.a.)$

**h)** $2\ (u.a.)$

**i)** $2\ (u.a.)$

**j)** $\frac{3\pi}{2}\ (u.a.)$

**k)** $\frac{9\pi}{2}\ (u.a.)$

**3.27.**

**a)** $\pi - \frac{3\sqrt{3}}{4}\ (u.a.)$ **b)** $---$

**c)** $---$ **d)** $---$

**e)** $---$ **f)** $---$

## B.7. Outros métodos de primitivação

**3.28.**

**a)** $P(x) = -\frac{1}{6}\cdot \ln|x| + \frac{3}{10}\cdot \ln|x-2| - \frac{2}{15}\cdot \ln|x+3| + C$

**b)** $P(x) = x - \ln|x| - \frac{3}{5}\cdot \ln|x-1| - \frac{1}{5}\cdot \ln|x^2+2x+2| - \frac{4}{5}\cdot \arctan(x+1) + C$

**c)** $P(x) = \frac{x^2}{2} + 2\cdot \ln|x| - \frac{1}{x} - 2\cdot \ln|x-1| + C$

**d)** $P(x) = x + \ln|x-1| - \frac{1}{2}\cdot \ln|4x^2+4x+4| - \sqrt{3}\cdot \arctan\left[\frac{1}{\sqrt{3}}\cdot(2x+1)\right] + C$

**e)** $P(x) = \frac{5}{2}\cdot \arctan(x) + \frac{x}{2\cdot(x^2+1)} + C$

**f)** $P(x) = \frac{\sqrt{2}}{2}\cdot \arctan\left[\frac{1}{\sqrt{2}}\cdot(x+1)\right] + \frac{1}{2}\cdot\frac{1}{(x^2+2x+3)} + C$

**g)** $P(x) = \frac{3}{16}\cdot \ln|x| - \frac{1}{8x} - \frac{3}{16}\cdot \ln|x-2| - \frac{5}{4}\cdot\frac{1}{(x-2)} - \frac{7}{8}\cdot\frac{1}{(x-2)^2} + C$

**h)** $P(x) = -\frac{1}{4}\cdot \ln|x+1| + \frac{1}{4}\cdot \ln|x-1| + \frac{1}{2}\arctan(x) + C$

**i)** $P(x) = -\arctan(x) + 2\cdot \arctan\left(\frac{x}{2}\right) + C$

**j)** $P(x) = \frac{\sqrt{2}}{8}\cdot \ln\left|\frac{x^2+1+\sqrt{2}x}{x^2+1-\sqrt{2}x}\right| + \frac{1}{2}\cdot\left[\frac{1}{\sqrt{2}}\arctan\left(\sqrt{2}x-1\right) + \frac{1}{\sqrt{2}}\arctan\left(\sqrt{2}x+1\right)\right] + C$

**k)** $P(x) = \ln|x^2+x+1| - \frac{2\sqrt{3}}{3}\cdot \arctan\left[\frac{2}{\sqrt{3}}\cdot\left(x+\frac{1}{2}\right)\right] + C$

**l)** $P(x) = \frac{3}{2}\cdot \ln|x^2+x+1| - \frac{\sqrt{3}}{3}\cdot \arctan\left[\frac{2}{\sqrt{3}}\cdot\left(x+\frac{1}{2}\right)\right] + C$

**m)** $P(x) = -\ln|x+1| + 3\cdot \ln|x+2| - 2\cdot \ln|x+3| + C$

**3.29.**

**a)** $P(x) = -\cos(x) + \frac{2 \cdot \cos^3(x)}{3} - \frac{\cos^5(x)}{5} + C$

**b)** $P(x) = -\frac{\sin^3(2x)}{48} - \frac{\sin(4x)}{64} + \frac{1}{16}x + C$

**c)** $P(x) = \frac{\sin^5(x)}{5} - \frac{\sin^7(x)}{7} + C$

**d)** $P(x) = \frac{\cos^5(x)}{5} - \frac{\cos^3(x)}{3} + C$

**e)** $P(x) = \frac{\tan^5(x)}{5} + \frac{\tan^3(x)}{3} + C$

**f)** $P(x) = \frac{\sec^3(x) \cdot \tan(x)}{4} - \frac{\sec(x) \cdot \tan(x)}{8} - \frac{1}{8}\ln|\sec(x) + \tan(x)| + C$

**g)** $P(x) = \frac{\sec^8(x)}{8} - \frac{\sec^4(x)}{2} + \frac{1}{2}\ln|\sec^2(x)| + C$

**h)** $P(x) = \frac{\operatorname{cosec}^5(x)}{5} - \frac{\operatorname{cosec}^7(x)}{7} + C$

**i)** $P(x) = \ln\left|\tan\left(\frac{x}{2}\right)\right| + C$

**j)** $P(x) = \frac{\sin(2x)}{4} - \frac{\sin(8x)}{16} + C$

**3.30.**

**a)** $P(x) = \frac{1}{2} \cdot x \cdot \sqrt{x^2+5} + \frac{5}{2} \cdot \ln\left|\frac{1}{\sqrt{5}} \cdot \left(x + \sqrt{x^2+5}\right)\right| + C$

$= \frac{1}{2} \cdot x \cdot \sqrt{x^2+5} + \frac{5}{2} \cdot \ln\left|\left(x + \sqrt{x^2+5}\right)\right| + K$

**b)** $P(x) = \frac{9}{2} \cdot \arcsin\left(\frac{x-3}{3}\right) + \frac{(x-3)}{2} \cdot \sqrt{x \cdot (6-x)} + C$

**c)** $P(x) = -\frac{4}{3} \cdot \sqrt{(2-x^2)} - \frac{x^2}{3} \cdot \sqrt{(2-x^2)} + C$

**d)** $P(x) = \ln\left|\frac{\sqrt{x^2-9}+x}{3}\right| - \frac{\sqrt{x^2-9}}{x} + C = \ln\left|\sqrt{x^2-9}+x\right| - \frac{\sqrt{x^2-9}}{x} + K$

**e)** $P(x) = \frac{1}{54} \cdot \operatorname{arcsec}\left(\frac{x}{3}\right) + \frac{\sqrt{x^2-9}}{18x^2} + C$

**f)** $P(x) = -\frac{1}{2} \cdot \ln\left|\sqrt{1-x^2}+1\right| + \frac{1}{2} \cdot \ln\left|\sqrt{1-x^2}-1\right| - \arcsin(x) + C$

**g)** $P(x) = \dfrac{x}{4 \cdot (4 + x^2)^{1/2}} + C$

**h)** $P(x) = \dfrac{1}{2} \cdot \arcsin\left(\dfrac{2x-1}{2}\right) + \left(\sqrt{3 + 4x - 4x^2}\right) \cdot \left(\dfrac{x^2}{3} - \dfrac{x}{12} - \dfrac{3}{8}\right) + C$

**i)** $P(x) = \sqrt{2} \cdot \ln|x| - \dfrac{\ln\left|2\sqrt{2} \cdot x \cdot \sqrt{2 + x - x^2} + x^2 + 4x\right|}{\sqrt{2}} + C$

**j)** $P(x) = \arcsin\left[\dfrac{1}{3} \cdot (x-1)\right] + C$

**k)** $P(x) = \ln\left|\sqrt{\dfrac{1}{41} \cdot (x^2 + 8x - 25)} + \dfrac{1}{\sqrt{41}} \cdot (x+4)\right| + C$

**3.31.**

**a)** $P(x) = \dfrac{\sqrt{5}}{5} \cdot \arctan\left[\dfrac{3}{\sqrt{5}} \cdot \tan\left(\dfrac{x}{2}\right) + \dfrac{\sqrt{5}}{5}\right] + K$

**b)** $P(x) = \dfrac{2}{\sqrt{3}} \cdot \arctan\left[\dfrac{1}{\sqrt{3}} \cdot \tan\left(\dfrac{x}{2}\right)\right] + K$

**c)** $P(x) = -x + \sqrt{2} \cdot \arctan\left[\dfrac{1}{\sqrt{2}} \cdot \tan(x)\right] + K$

**d)** $P(x) = \dfrac{1}{5} \cdot \ln\left|\left(3 \cdot \tan\left(\dfrac{1}{2}x\right) - 1\right)\right| - \dfrac{1}{5} \cdot \ln\left|\left(\tan\left(\dfrac{1}{2}x\right) + 3\right)\right| + K$

**e)** $P(x) = \dfrac{1}{5}\left[\ln\left|\dfrac{8}{5} + \dfrac{4}{5} \cdot \tan\left(\dfrac{x}{2}\right)\right| - \ln\left|-\dfrac{2}{5} + \dfrac{4}{5} \cdot \tan\left(\dfrac{x}{2}\right)\right|\right] + K$

**f)** $P(x) = -\dfrac{1}{2} \cdot \left[\dfrac{\tan^2\left(\dfrac{x}{2}\right)}{2} - \ln\left|\tan\left(\dfrac{x}{2}\right)\right|\right] + K$

**g)** $P(x) = -\dfrac{1}{3} \cdot \ln\left|\left(2 \cdot \tan\left(\dfrac{x}{2}\right) - 1\right)\right| + \dfrac{1}{3} \cdot \ln\left|\left(\tan\left(\dfrac{x}{2}\right) - 2\right)\right| + K$

**h)** $P(x) = 2 \cdot \arctan[\sin(x)] - \sin(x) + K$

**i)** $P(x) = 4 \cdot \left[\dfrac{1}{3}(1 + \sin(x))^{\frac{3}{2}} - \sqrt{1 + \sin(x)}\right] + K$